Ex.No: 18 Date:

Create a Simple Network using Cisco Packet Tracer

AIM:

To design and build a simple Local Area Network (LAN) in Cisco Packet Tracer by connecting end devices through a switch, assigning IP addresses, and verifying end-to-end connectivity with the ping command.

SOFTWARE REQUIRED:

· A computer connected to a live network (Ethernet or Wi-Fi). · Cisco Packet Tracer (version 8.x recommended · Basic knowledge of IPv4 addressing and subnet masks.

THEORY:

Cisco Packet Tracer is a network simulation tool that lets you build and test networks without physical hardware. A simple LAN connects several end devices (PCs) to a central switch. A switch forwards frames using MAC addresses and creates one broadcast domain, so all devices attached to it — if given IP addresses in the same subnet — can communicate directly. In this experiment three PCs are joined to a single switch on the network 192.168.1.0/24 and connectivity is proven with ping, which sends ICMP echo requests and waits for replies.

Addressing plan

Device Interface IP Address Subnet Mask

PC0 FastEthernet0 192.168.1.1 255.255.255.0

PC1 FastEthernet0 192.168.1.2 255.255.255.0

PC2 FastEthernet0 192.168.1.3 255.255.255.0

PROCEDURE:

1. Open Cisco Packet Tracer. You are presented with a blank logical workspace (the large white area).

2. Add a switch: from the bottom-left device box choose Network Devices → Switches, pick the 2960 model, and click on the workspace to drop it.

3. Add three PCs: choose End Devices, select PC, and place PC0, PC1 and PC2 around the switch.

4. Connect the devices: choose the Connections (lightning-bolt) icon and pick Copper Straight-Through cable. Click a PC, choose its FastEthernet0 port, then click the switch and choose a free FastEthernet port. Repeat for all three PCs.

5. Wait until the link lights turn green (a few seconds), indicating the ports are up.

6. Configure IP addresses: click PC0 → Desktop tab → IP Configuration. Select Static and enter IP 192.168.1.1 and subnet mask 255.255.255.0. Close the window.

7. Repeat step 6 for PC1 (192.168.1.2) and PC2 (192.168.1.3), all with mask 255.255.255.0.

8. Test connectivity: on PC0 → Desktop → Command Prompt, run the ping command to another PC.

9. PC> ping 192.168.1.2 PC> ping 192.168.1.3

10. A successful ping shows four replies with round-trip times. Repeat from other PCs to confirm every device can reach every other device.

11. Optional: switch to Simulation mode (bottom-right) and re-run a ping to watch the ICMP packet animate hop-by-hop from source to switch to destination.

12. Save the file via File → Save As with a .pkt extension.

SAMPLE OBSERVATION TABLE:

C:\> ping 192.168.1.2

Pinging 192.168.1.2 with 32 bytes of data: Reply from 192.168.1.2: bytes=32 time<1ms TTL=128 Reply from 192.168.1.2: bytes=32 time<1ms TTL=128 Reply from 192.168.1.2: bytes=32 time<1ms TTL=128 Reply from 192.168.1.2: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.2: Packets: Sent = 4, Received = 4, Lost = 0 (0% loss), If the first request “times out”, it is normal — it is caused by ARP resolving the MAC address; subsequent replies confirm the link is working.

RESULT: Thus a simple LAN of three PCs connected through a 2960 switch was built in Cisco Packet Tracer. IP addresses were assigned in the 192.168.1.0/24 network, and end-to-end connectivity between all devices was successfully verified using the ping command.